Clara AWS Cost Optimization Executive Report

1. Executive Summary

Clara operates a multi-account AWS environment supporting a fast-growing microservices architecture with a current monthly cloud spend of **$250,000 USD**. The largest cost drivers are **EC2, EBS, RDS, S3, EKS, and Lambda**, with EC2 representing **40%** of total monthly spend.

The analysis identified optimization opportunities related to underutilized compute resources, unattached EBS volumes, excessive snapshot retention, high S3 Standard storage usage, and static EKS node capacity. The proposed plan estimates approximately **$70,000 USD in monthly savings**, reducing AWS spend from **$250,000 USD to $180,000 USD**, equivalent to a **28% reduction**.

This recommendation follows a conservative FinOps approach: improve visibility, remove waste, rightsize infrastructure, optimize storage lifecycle, and apply commitment-based discounts only after workload patterns are validated.

2. Current AWS Cost Baseline

Service	Monthly Cost	% Spend	Key Observation
EC2	$100,000	40%	500 instances; 60% are c5.xlarge; average CPU utilization is 30%.
EBS	$50,000	20%	200 unattached volumes and excessive snapshot retention.
RDS	$37,500	15%	Future opportunity for reserved capacity and storage review.
S3	$25,000	10%	50 TB stored; 80% remains in S3 Standard.
EKS	$25,000	10%	3 clusters with 10 c5.xlarge nodes each; static node capacity.
Lambda	$12,500	5%	Smaller opportunity; memory and duration tuning require profiling.
Total	**$250,000**	**100%**	Current AWS monthly baseline.

The cost profile shows that Clara's main issue is not a single expensive service, but insufficient governance across compute, storage, Kubernetes capacity, snapshots, tagging, and lifecycle automation.

3. Prioritized Recommendations

Priority	Recommendation	Estimated Monthly Savings	Estimated Impact	Implementation Focus
1	EC2 and EKS Compute Optimization	$42,500	17%	Rightsizing, Auto Scaling, EKS node optimization, Savings Plans after validation.
2	EBS Volume and Snapshot Optimization	$17,500	7%	Delete validated unattached volumes, migrate eligible workloads to gp3, apply snapshot lifecycle policies.
3	S3 Storage Lifecycle Optimization	$10,000	4%	Use S3 Storage Lens, Intelligent-Tiering, lifecycle rules, and cleanup of incomplete multipart uploads.
Total	**Primary Optimization Plan**	**$70,000**	**28%**	**Conservative savings estimate based on validated opportunities.**

Recommendation 1: EC2 and EKS Compute Optimization

EC2 and EKS are the largest optimization areas. Clara currently runs 500 EC2 instances with average CPU utilization of only 30%, suggesting that part of the fleet may be oversized or running continuously without matching workload demand.

Recommended actions include using AWS Compute Optimizer and CloudWatch metrics to identify underutilized instances, rightsizing based on workload behavior, applying Auto Scaling policies, optimizing EKS node groups using Cluster Autoscaler or Karpenter, using mixed instance types, and applying Savings Plans only after the stable compute baseline is validated.

Recommendation 2: EBS Volume and Snapshot Optimization

EBS represents 20% of Clara's monthly AWS spend. The presence of 200 unattached EBS volumes and approximately 10 snapshots per in-use volume indicates direct storage waste and weak lifecycle governance.

Recommended actions include deleting validated unattached volumes, migrating suitable workloads to gp3, implementing snapshot lifecycle policies, defining retention standards by workload criticality, and enforcing tagging for ownership and cost allocation.

Recommendation 3: S3 Storage Lifecycle Optimization

Clara stores 50 TB of data in S3, with 80% in the Standard storage class. This indicates potential savings through storage tiering and lifecycle policies.

Recommended actions include analyzing access patterns with S3 Storage Lens, moving infrequently accessed data to S3 Intelligent-Tiering, applying lifecycle transitions to Standard-IA or Glacier tiers where appropriate, and cleaning incomplete multipart uploads and expired objects.

4. Estimated Savings Summary

Metric	Amount
Current Monthly AWS Spend	$250,000
Estimated Monthly Savings	$70,000
Optimized Monthly AWS Spend	$180,000
Estimated Reduction	28%
Estimated Annualized Savings	$840,000

The executive model intentionally excludes immediate savings from RDS and Lambda. These services are treated as future opportunities requiring additional workload validation, capacity analysis, and profiling before applying optimization targets.

5. Implementation Plan

Phase	Timeline	Main Actions	Expected Outcome
Phase 1: Visibility and Governance	Weeks 1–2	Enable or validate Cost Explorer, Cost and Usage Report, AWS Budgets, Cost Anomaly Detection, tagging standards, and cost baselines by account, service, workload, and owner.	Reliable baseline and governance model.
Phase 2: Waste Removal and Rightsizing	Weeks 3–6	Remove validated unattached EBS volumes, apply snapshot lifecycle policies, rightsize EC2, optimize EKS node groups, and apply S3 lifecycle policies.	Direct savings from waste removal and capacity optimization.
Phase 3: Rate Optimization and Continuous FinOps	Ongoing	Apply Savings Plans or Reserved Instances after rightsizing, review RDS opportunities, create monthly cost reviews, dashboards, budgets, and anomaly controls.	Sustained savings and prevention of cost regression.

Production changes should be phased, beginning with development and staging workloads before production. Any deletion or rightsizing action must require owner validation, rollback planning, and monitoring of performance indicators.

6. Conclusion

Clara can reduce AWS monthly spend by approximately **$70,000 USD**, or **28%**, through a focused FinOps plan targeting EC2/EKS rightsizing, EBS cleanup, snapshot lifecycle management, and S3 storage class optimization.

The recommendations are feasible, measurable, and aligned with operational reliability. The plan avoids aggressive one-time cuts and instead establishes repeatable cost governance through visibility, ownership, lifecycle policies, Rightsize controls, and continuous FinOps reviews.